

RACTF 2020 Crypto - 02

deuterium — 2020-06-09

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/ractf/crypto/02/2020-06-09-RACTF-2020-02.html>

Abstract

CTF writeup: The alphabet keeps changing shifts in RACTF 2020's 02. Identify Vigenere and let dCode fish out the key.

02

Progressing from 01 we get to 02 and are presented with the following ciphertext

```
KFCHT QXXKR FSAHX IEIYP GYZRX
YXCKK OKYPG YLNIX BQRFU WFKEH
LNYGC VBDGT NVIMF NJJLV HJEJY
PGZFO IKQTL KBJKW TXNEH FVEHD
PQJBG MYEYW IPLRC YNWPM YEKNV
CEKRF SAHXI MFDVG XUTTQ MRXIF
TWUGW ZNUJZ UHEBJ FKWLK MDECT
BTHGF VMTGP JFZUM FHFAM UNJPN
HQQGJ AGTCV MYEVZ IPMZT NJAQY
DOSJG DXZNL RWXXU AWTCP WZFDT
CCKVA AFQNT SLJBM ETEAW FVIXR
MJJBK GXPTN VVTED HTURE VTJYP
GWVAQ DWWKJ DTSVK X
```

Expecting the level to increase slightly, the first guess is Vignere

Without wasting time lets go to dcode.fr to see if it works, and it confidently spews out the plaintext

```
ROCCO IVETR ANSFE REDTH EFIRS
TPART OFTHE FUNDS TOYOU RACCO
UNTBU TIMGO INGTO NEEDT OSEET
HEGOO DFIRS TBEFO REWEC ANCOM
PLETE THETR ANSAC TIONT HEFIN
ALTRA NSFER MAYNE EDTOB EPERF
ORMED INPER SONBE ACUSE MYZUR
ICHBA NKAPP EARST OHAVE SUSPI
CIONS ABOUT THEQU ANTIT IESOF
MONEY BEING MOVED AROUN DIFYO
UAREA VAILA BLEWE CANAR ANGEA
MEETI NGPOI NTAND COMPL ETETH
EDEAL YOURS DONNI E
```

The flag is the location of the bank

and the key TRAFFIC

rearranging the spaces, we get

ROCCO I VE TRANSFERED THE
FIRST PART OF THE FUNDS TO YOU RACCO
UNT BUT I M GOING TO NEED TO SEE
THE GOOD FIRST BEFORE WE CAN COMPLETE THE
TRANSACTION THE FINAL TRANSFER MAY NEED TO BE
PERFORMED IN PERSON BEACUSE MY ZURICH BANK APPEARS TO
HAVE SUSPICIONS ABOUT THE QUANTITIES OF
MONEY BEING MOVED AROUND IF YOU ARE
AVAILABLE WE CAN ARANGE A
MEETING POINT AND COMPLETE THE DEAL YOURS DONNIE

And we can see people talking about **ZURICH** which is the place we seek as the location of the bank

For a better analysis, check [this video](#) by @hyperreality

| Interactive controls are omitted from this PDF. See the canonical HTML version.